

System Requirements Documentation

Woodworking Order Management System

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Introduction

The project I am creating is a Woodworking Order Management System for a small custom woodworking business. The business currently handles all its daily order tracking manually. This includes writing down customer information, keeping track of custom project requests, updating order progress, recording payment information, and due dates of projects. This manual process worked when the business was just starting out and had only a few orders, it has become harder to manage as the business has grown. The main purpose of this new system is to give the woodworking business a simple and organized way to manage customer orders. Instead of relying on paper notes, scattered files, or memory. The system will allow the user to enter order information, save it, view it later, update the order status, and track payment details. The system will not be complicated. It is designed for a small business that needs an effective tool to improve organization, reduce mistakes, and keep track of customer orders.

This project will be developed using HTML, CSS, and JavaScript in Notepad++. The system will run in a web browser and use local Storage browser to save order information. The project does not require a separate database server. The data will still be structured in a way that supports customer and order management, but it will be stored locally in the browser for this version of the project. The final system will display many important software engineering concepts, including system implementation, data model design, user interface design, software testing, error handling, debugging, performance planning, and post-delivery maintenance. The goal is to show how a simple software system can help a real business solve a problem by replacing an outdated manual process with a more organized digital system.

Customer Problem Statement and System Requirements

The woodworking business in this project is a small custom shop that makes products like tables, cabinets, cutting boards, and many other custom wood projects. The business is still using a manual process to keep track of its work. Customer information, order details, payments, and project due dates are written down on paper or stored in different places. The main problem is that the business does not have one place to manage all the customer orders. For example, the customer's contact information might be in one place, the project description might be somewhere else, and the payment information might be tracked separately. This makes it very hard for the owner and employees to quickly know what needs to be worked on, what has already been paid, and which orders are still unfinished.

Another big issue is that manual tracking can lead to many mistakes. If the owner calculated balances by hand or looked through notes to find an order status, there is a much higher chance that something would be messed up. A missed payment, wrong balance, or forgotten due date could cause confusion with the customer and make the business look unprofessional. The business needs a better way to track the progress of each woodworking order as well. A project goes through several stages, like requested, approved, in progress, completed, and delivered. Without a system to track these stages, the owner may have to rely too much on memory or handwritten notes. This would lead to delays. The proposed Woodworking Order Management System will help solve these problems by giving the business a simple web-based tool for managing orders. The system will let the user enter customer information, create a woodworking order, track the estimated cost and amount paid, calculate remaining balance, update the order status, and view saved orders in one place. The goal is not to create an overcomplicated system, but to create something that's useful.

Functional Requirement Specification

The Woodworking Order Management System will include the main functions that are needed to help a small woodworking business manage its customer orders. The system will let user add a new order by entering the customer's name, phone number, email address, project type, project description, estimated cost, amount paid, due date, and current order status. This keeps all the important information about the order in one place. Since the project will be built with HTML, CSS, and JavaScript, the system will use local Storage browser to save order information. This will allow the saved orders to still be available after the page is refreshed.

The system will also display all saved orders in a clear list or table so the user can quickly review what work needs to be done. The user will be able to see important details like the customer's name, type of project, due date, payment information, and order status. The system will let the user update an order's status as the project moves through stages such as requested, approved, in progress, completed, and delivered. This will help the owner keep track of where each project is in the cycle. The system will calculate the remaining balance by subtracting the amount paid from the estimated cost, which will help reduce payment tracking mistakes.

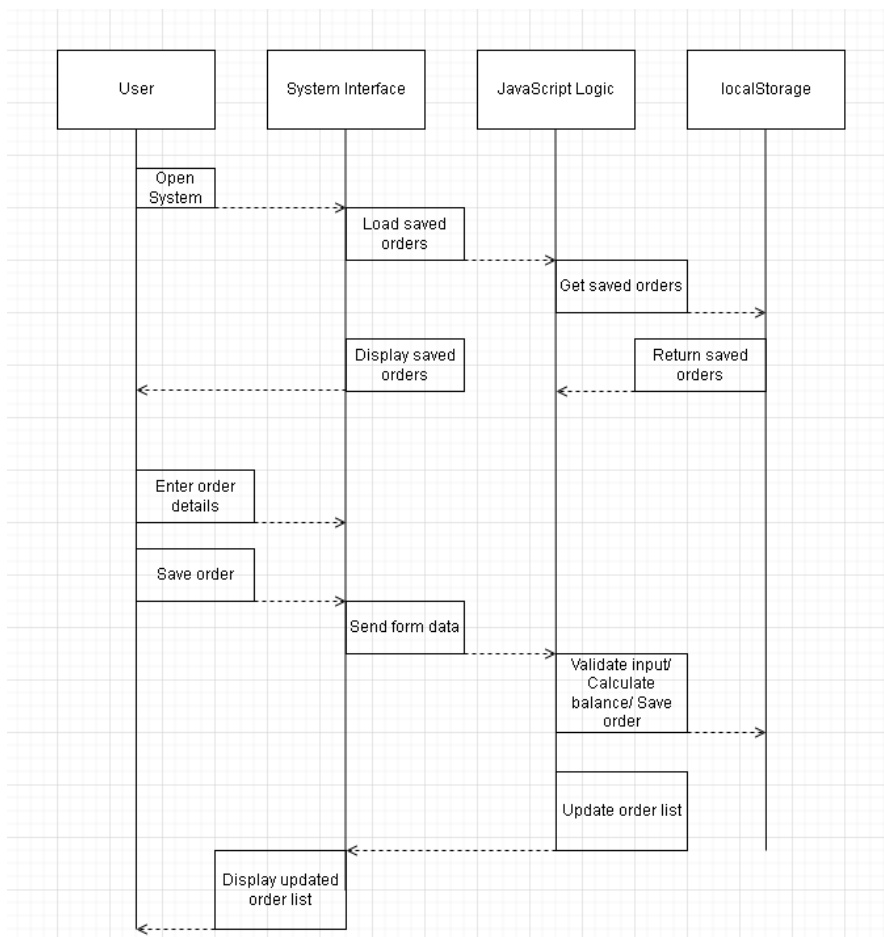
Another important function of the system is letting users delete an order if it was entered by mistake, canceled, or no longer needed. The system will include basic validation and error handling. These functional requirements support the main goal of the project, which is to replace a manual order tracking process with a simple digital system that is easier to organize, update, and maintain.

System Sequence Diagram

The system sequence diagram shows how the user will interact with the Woodworking Order Management System. The main user will be the woodworking business owner or employee who is entering and managing orders. The process starts when the user opens the system in the browser. Then the user can enter customer information, project details, payment information, and the order status. After the user submits the form, the system checks that the required information is entered right.

If the information is valid, the system calculates the remaining balance by subtracting the amount paid from the estimated cost. Then the system saves the order information in browser localStorage. When the order is saved, the system updates the order list on the screen so the user can view the new order with the other saved orders. The user can update the order status later or delete an order if it was entered by mistake or is no longer needed.

The diagram below represents the basic sequence for adding and saving a woodworking order. It shows the main communication between the user, the system interface, the JavaScript logic, and localStorage.

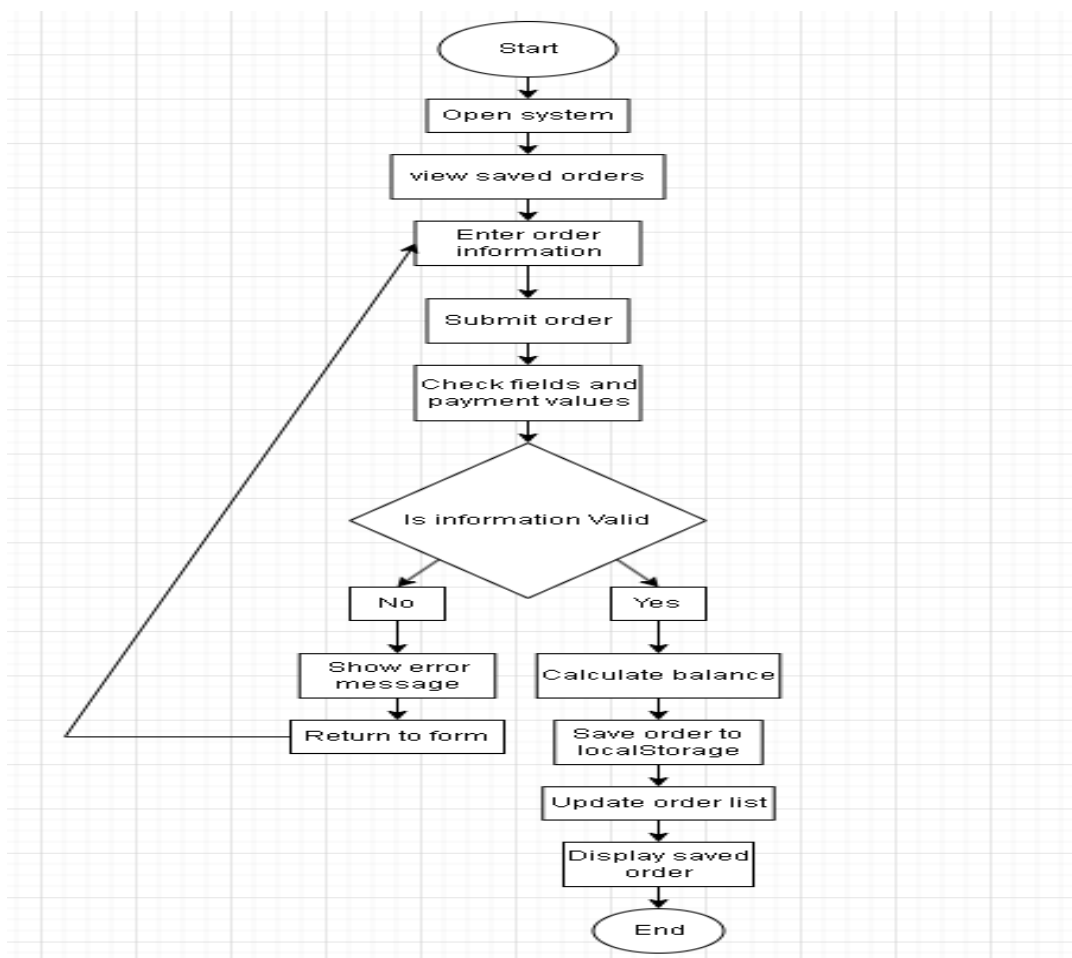


Activity Diagram

The activity diagram shows the main steps the user will follow when using the Woodworking Order Management System. The process starts when the user opens the system in the browser. From there, the user can choose to add a new woodworking order, view current orders, update an order status, or delete an order. The main activity for this system is adding and saving an order because that is the main problem the business is trying to fix.

When the user adds a new order, they will enter the customer information, project details, estimated cost, amount paid, due date, and order status. The system will then check that the required fields are filled in and that the payment information is entered correctly. If something is missing or wrong, the system will show an error message. If everything is entered right, the system will calculate the remaining balance, save the order to localStorage, and update the order list on the screen.

This activity diagram helps show how the system changes the current manual process into a more organized digital process. The user can follow a clear set of steps in the system.



User Interface Specification

The user interface for the Woodworking Order Management System will be designed to be simple and easy to use. Since the system is meant for a small woodworking business, the screen should not be crowded or difficult to understand. The main page will include a title at the top, a form for entering customer and order information, and a section that displays the saved orders. The form will include fields for the customer's name, phone number, email address, project type, project description, estimated cost, amount paid, due date, and order status. These fields are important because they collect the main information that the business needs to keep track of each woodworking order.

The interface will include buttons that allow the user to save an order, update an order status, and delete an order if needed. After an order is saved, it will appear on a table or list on the page. This list will show the most important details, such as customer name, project type, due date, balance due, and order status. This will make it easier for the owner or employee to quickly see what projects are active and what still needs to be finished. The goal is to make the system feel like a useful order tracker.

The design will use basic HTML, CSS, and JavaScript. HTML will be used to build the structure of the form and order list, CSS will be used to make the page easier to read, and JavaScript will control the actions such as saving, displaying, updating, and deleting orders. The interface will show simple error messages if required information is missing or if payment values are entered wrong. The user interface will focus on being clear, organized, and easy for the business owner to use without needing advanced computer knowledge.

Implementation Planning

The implementation plan for the Woodworking Order Management System will start with setting up the project folder in Notepad++. The main files will include an HTML file for the page structure, a CSS file for the design, and a JavaScript file for the system logic. The project will include separate folders for the code, testing files, data collection, and documentation. This will keep the project organized and make it easier to find each part of the final submission.

The first part of the build will focus on creating the main user interface. This will include the page title, the order entry form, and the section where saved orders will be displayed. After the interface is created, the JavaScript will be added to make the system work. The JavaScript will collect the form information, check that the required fields are filled in, calculate the remaining balance, save the order to localStorage, and display the saved

orders on the page. Once the basic add order feature works, the update status and deleted order features will be added.

After the main features are working, the system will be tested to make sure it behaves the way it is supposed to. Testing will include checking that orders can be added, saved, viewed, updated, and deleted. It will include checking that the balance is calculated correctly and that the system shows an error message when required information is missing.

System Architecture and System Design

The Woodworking Order Management System will use simple client-side architecture. This means the system will run in the user's web browser instead of using a separate server or database. The project will be made with three main files: an HTML file, a CSS file, and a JavaScript file. The HTML file will create the structure of the page, including the order form and the saved order list. The CSS file will control the layout, colors, spacing, and visual appearance of the system. The JavaScript file will handle the main system logic, such as collecting order information, checking for missing fields, calculating the balance due, saving orders, displaying orders, updating order status, and deleting orders.

The system design is based on keeping the project easy to use and easy to understand. When the user opens the system, they will see the main order form and a section for saved orders. The user will enter the customer and project details into the form. When the order is submitted, JavaScript will check the information, calculate the remaining balance, and save the order to browser localStorage. The saved order will then appear on the page so the user can view it later. Since the system uses localStorage, the saved orders will still be there after the page is refreshed, but the data will stay on the same browser and computer.

This design fits the needs of the woodworking business because the business does not need a complicated system to solve the current problem. The main goal is to replace paper notes and scattered information with a small digital tool that keeps orders organized. The system architecture is simple, but it still supports the important parts of the project, including the user interface, data storage, order management, payment tracking, and basic error handling. This setup makes the project easier to test and maintain because each file has a clear purpose.

Algorithms and Data Model Implementation

The main algorithm for the Woodworking Order Management System will focus on adding, saving, and displaying customer orders. When the user enters a new order, the system will first collect the information from the form. This will include the customer's name, phone number, email, project type, project description, estimated cost, amount

paid, due date, and order status. The system will check that the required fields are filled in and that the cost and payment amounts are valid numbers. If the information is right, the system will calculate the remaining balance by subtracting the amount paid from the estimated cost. Then the order will be saved and displayed in the order list.

The data model will be simple because this project does not use a separate database. Each order will be stored as a JavaScript object and saved in browser localStorage. Each order record will include an order ID, customer name, phone number, email address, project type, project description, estimated cost, amount paid, balance due, due date, order status, and date created. The order ID will help keep each order separate from each other, even if two customers have similar information. The system will treat the saved orders like a small list of records that can be added to, displayed, updated, or deleted.

This data model fits the project because the woodworking business mainly needs a simple way to organize order information. The customer information, project details, and payment information are all connected to one order. For this system version, that is enough to show how the business can move from paper tracking to a digital system. If the system was expanded in the future, the data model could be changed into separate tables for customers, orders, and payments.

User Interface Design and Implementation

The user interface design for the Woodworking Order Management System will focus on being simple, organized, and easy to follow. The main page will have a heading at the top, so the user knows they are in the order management system. Under that section, there will be an order form where the user can enter the customer information, project details, payment amounts, due date, and order status. The form will be kept in one main section, so the user does not have to move between several different pages to create an order.

The saved orders will be shown below the form in a table or list format. This will let the user quickly see the important details for each order, such as the customer's name, project type, due date, balance due, and status. The design will include buttons for saving a new order, updating an order status, and deleting an order. The page will use simple styling with CSS so the form and order list are easier to read. The design is not meant to be flashy. The main goal is to make the information clear and useful for the woodworking business.

The interface will be implemented using HTML, CSS, and JavaScript. HTML will create the form fields, labels, buttons, and order display area. CSS will be used to organize the layout and make the page look clean. JavaScript will connect the interface to the system logic by

reading the form values, checking the input, saving the order to localStorage, and updating the order list on the screen.

Main System Screen

Woodworking Order Management

C:\Users\jdver\OneDrive\Documents\INFO-C451\Final_Project\WoodWorking_Order_System\1_Code\index.html

☆ ☆ 👤 ⋮

Woodworking Order Management System

Manage customer orders, payments, and project status in one place.

Add New Order

Customer Name

Phone Number

Email

Project Type

Table, cabinet, cutting board, etc.

Project Description

Estimated Cost

Amount Paid

Due Date

mm / dd / yyyy

Order Status

Select Status

Save Order

Saved Orders Table with Order Summaries

Order Summary

Total Orders: 3

Total Balance Due: \$1750.00

Saved Orders

Customer	Project	Due Date	Status	Balance Due	Actions
Justin Verrick	Dining Table	2026-06-01	In Progress	\$1500.00	Delete
John Doe	Bookshelf	2026-05-03	Completed	\$0.00	Delete
Amanda Smith	Cutting Board & Coaster set	2026-06-02	Requested	\$250.00	Delete

Error Message for Missing Fields

Woodworking Order Management System

Manage customer orders, payments, and project status in one place.

Add New Order

Customer Name	A
Phone Number	V
Email	
Project Type	Table, cabinet, cutting board, etc.
Project Description	
Estimated Cost	
Amount Paid	
Due Date	mm / dd / yyyy
Order Status	Select Status

Please fill in all required fields before saving the order.

Save Order

Designs of Tests

The testing for the Woodworking Order Management System will focus on making sure the main features work right. The system is built to help a woodworking business manage customer orders; the most important tests will be based on adding an order, saving the order, displaying saved orders, updating the order status, deleting an order, and calculating the balance due. The goal of the testing is to make sure the system works the way the user expects and does not lose or display incorrect information.

Unit testing will be used to check smaller parts of the system. For example, one test will check that the balance due is calculated right by subtracting the amount paid from the estimated cost. Another test will check that required fields can't be left blank when adding a new order. Another test will check that payment values are valid numbers and not negative amounts. These tests are important because small errors in the system could cause bigger problems later, even with payment tracking.

Integration testing will be used to check that the different parts of the system work together. For example, when the user fills out the order form and clicks the save button, the system should validate the information, calculate the balance, save the order to

localStorage, and then display it on the page. Another integration test will check that an order can be updated or deleted after it has been saved. These tests will help make sure the form, JavaScript logic, localStorage, and order display area are all working together.

Error Handling and Debugging

Error handling for the Woodworking Order Management System will focus on preventing common mistakes when the user is entering order information. The system will check that important fields are not left blank, such as the customer's name, project type, estimated cost, amount paid, due date, and order status. It will check that the estimated cost and amount paid are entered as valid numbers. If the user enters something wrong or forgets a required field, the system will show a simple error message so the user knows what needs to be fixed before the order can be saved.

The system will also help prevent payment tracking errors. For example, the amount paid should not be a negative number, and the balance due should be calculated automatically by subtracting the amount paid from the estimated cost. This reduces the chance of the user making a mistake by calculating the balance manually. If an order is entered by mistake or no longer needed, the user will be able to delete it from the system. These basic error handling features will make the system more reliable and easier on the users.

Debugging will be done while building and testing the system. Since the project uses HTML, CSS, and JavaScript, I will use the browser's developer tools to check for problems in the console and test whether the buttons, form fields, and localStorage are working correctly. I will test the system by adding sample orders, refreshing the page to make sure the data is still saved, updating order statuses, deleting orders, and checking that error messages appear when something is entered incorrectly. This will help find issues early on and make sure the system works right.

Collaboration and Code Integration

The Woodworking Order Management System will be completed as an individual project, so the collaboration part will mainly focus on keeping the project organized and making sure each part of the code works together correctly. The project will be separated into different files so the code is easier to manage. The HTML file will handle the page structure, the CSS file will handle the design, and the JavaScript file will handle the system logic. Keeping the files separate will make it easier to find problems and make changes without having to search through one large file.

Code integration will happen as each part of the system is added. The first part will be the basic HTML form and order display area. After that, CSS will be added to make the layout easier to read. Then JavaScript will be connected to the form so the system can collect

order information, validate the input, calculate the balance, save the order to localStorage, and display the saved orders on the page. Each new feature will be tested before moving on to the next one.

Even though this is not a group project, the files will still be organized in a way that would make sense if someone else had to review or update the system later. Clear file names, simple comments, and a folder structure for code, testing, data collection, and documentation will help keep the project understandable.

Performance Optimization Plan

The Performance Optimization Plan for the Woodworking Order Management System will focus on keeping the system simple and fast. Since the project will run in a browser and use HTML, CSS, JavaScript, and localStorage, it should not need a lot of resources to work correctly. The system is only meant to manage woodworking orders for a small business, so the code will stay lightweight and not include extra features that are not needed. Keeping the layout simple and the files organized will help the page load quickly and make the system easier to use.

The JavaScript will be written so it only updates the parts of the page that need to change. For example, when a new order is added, the system will save the order and refresh the order list instead of reloading the whole page. The order records will be stored in a simple structure, which will make it easier for the system to read, display, update, and delete orders. The project uses localStorage, the amount of saved data should stay reasonable. This system is not meant to hold thousands of records, but it should work well for a small woodworking shop with a normal number of active orders.

Performance will be checked during testing. I will test the system by adding multiple sample orders, refreshing the page, updating statuses, and deleting records to make sure the system still responds the way it's designed to. I will check that the page is easy to read and that buttons respond without delay. If the system starts to feel slow or cluttered, I can improve it by simplifying the layout, removing unnecessary code, and keeping the order display clean. The main goal is for the system to load quickly, respond correctly, and remain easy for the business to use.

Maintenance and Support Strategy

The Maintenance and Support Strategy for the Woodworking Order Management System will focus on keeping the system easy to update and easy to fix. The project is built with HTML, CSS, and JavaScript, future changes can be made by editing the correct file. For example, changes to the page structure would be made in the HTML file, design changes would be made in the CSS file, and changes to the system logic would be made in the

JavaScript file. Keeping the files separated will make maintenance easier because each file has its own unique purpose.

Support for the system will include checking for common issues, such as orders not saving correctly, buttons not responding, or balance calculations being wrong. If a problem happens, the browser's developer tools can be used to check for errors in the console. The system can be tested with sample orders to make sure localStorage is saving and loading the data right. Since the system stores data in the browser, the user should understand that clearing browser data could remove saved orders. Because of this, future updates could include a backup option.

Long-term maintenance would include reviewing the system as the woodworking business grows. If the business starts handling more customers or needs more features, the system could be improved later. Possible updates could include search options, separate customer records, payment history, or an actual separate database. For this version of the system, the goal is to keep it simple and useful.

Reference

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